

Influence of Hydrogen Enrichment of Methane on Laminar Flame Speed and NO_x Emissions

Computational Methods in Combustion

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1 Introduction

The global energy sector is under increasing pressure to reduce greenhouse gas emissions and transition toward sustainable energy carriers. Natural gas, composed predominantly of methane, currently plays a central role in power generation and industrial heating owing to its relatively low carbon intensity compared with coal or oil. However, even complete combustion of methane releases CO_2 , and the transition to a truly low-carbon energy system requires either carbon capture or a shift toward carbon-free fuels.

Hydrogen is widely regarded as a key element of this transition. It releases no CO_2 upon combustion, has a very high energy content per unit mass, and can be produced from renewable electricity via electrolysis (so-called green hydrogen). A pragmatic near-term pathway to decarbonisation is the blending of H_2 into the existing natural gas grid—a strategy already tested at the distribution scale in several countries. Such blending enables incremental emission reductions while the infrastructure, safety standards, and economic frameworks for pure hydrogen supply chains mature.

From a combustion-engineering perspective, however, hydrogen addition to methane significantly alters flame behaviour. The high diffusivity and low ignition energy of H_2 can improve flame stability, but also raise safety concerns related to flashback and detonation risk. Changes in laminar flame speed, radical chemistry, and adiabatic flame temperature all directly affect burner design, emission formation, and operability limits. A quantitative, mechanistically grounded understanding of these effects is therefore essential for engineers tasked with adapting existing combustion equipment to hydrogen-enriched fuels.

This report addresses this need through a computational parametric study. The laminar flame speed S_u , maximum flame temperature T_{max} , NO emissions, radical peak concentrations, and spatial flame structure are evaluated as functions of hydrogen volume fraction in the CH_4/H_2 fuel blend.

2 Literature Review

The combustion characteristics of hydrogen-enriched natural gas (HENG) have been the subject of extensive research over the past two decades. The following paragraphs summarise the principal findings relevant to the present study.

Laminar flame speed. It is well established that the laminar burning velocity of $\text{CH}_4/\text{H}_2/\text{air}$ mixtures increases monotonically with hydrogen fraction. Huang et al. [1] and Halter et al. [2] measured this increase experimentally and attributed it to both the significantly higher intrinsic burning velocity of pure hydrogen (≈ 200 cm/s at stoichiometry) and to enhanced chain-branching kinetics that H_2 introduces into the methane oxidation pathway. Di Sarli and Di Benedetto [3] confirmed through numerical simulation that even moderate H_2 additions (below 30 %) produce a disproportionate increase in S_u relative to what a linear blending rule would predict.

Radical pool and chemical kinetic enhancement. The kinetic acceleration of HENG flames is closely linked to the enrichment of the radical pool, particularly OH, H, and O. Hydrogen addition provides additional H atoms via thermal decomposition, which then participate in the key chain-branching reaction $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$. This positive feedback loop greatly amplifies the overall reactivity of the mixture beyond the contribution of H_2

as a pure energy source. Studies using sensitivity analysis [4,5] consistently identify the $\text{H} + \text{O}_2$ reaction as the most sensitive step, and confirm that radical-pool enhancement is the dominant kinetic mechanism at low-to-moderate H_2 blending ratios.

Adiabatic flame temperature and NO_x. Theoretical considerations and measurements agree that H_2 addition raises the adiabatic flame temperature at stoichiometric conditions, primarily because the lower carbon-to-hydrogen ratio reduces heat capacity while the combustion enthalpy per unit volume remains high. This temperature increase has a strong positive effect on thermal NO formation via the extended Zeldovich mechanism [6], and numerous studies [7,8] report elevated NO_x emissions in HENG flames proportional to the temperature increase. The consensus is that the NO_x penalty is a significant engineering challenge that must be addressed through staged combustion, lean premixing, or exhaust gas recirculation.

Flame structure. Spatially resolved measurements and simulations show that hydrogen enrichment decreases the flame thickness—defined as the width of the preheat zone—due to the higher thermal diffusivity of H_2 and the more compact reaction zone. This results in a narrower, more intense heat-release profile and influences flame–turbulence interaction in practical burners.

3 Model Description

3.1 Software and Kinetic Mechanism

All simulations were performed using Cantera 3.2 [9], an open-source object-oriented software package for chemical kinetics and reacting-flow calculations. The one-dimensional freely propagating premixed flame model (`ct.FreeFlame`) was employed, which solves the steady-state conservation equations for mass, species, and energy on an adaptive grid using the Newton method with automatic Jacobian evaluation.

The GRI Mech 3.0 mechanism [10] (`gri30.yaml`) was used for all simulations. It contains 53 species and 325 elementary reactions, covering methane oxidation, C_2 chemistry, and the full NO_x formation sub-mechanism including the thermal (Zeldovich), prompt (Fenimore), and N_2O pathways. This makes it the standard choice for studying NO_x formation in methane-based premixed flames.

3.2 Flame Configuration and Boundary Conditions

The computational setup is summarised in Table 1.

Table 1: Simulation parameters.

Parameter	Symbol	Value
Unburned gas temperature	T_u	300 K
Pressure	P	1 atm (101 325 Pa)
Equivalence ratio	ϕ	1.0
H_2 fraction in fuel	x_{H_2}	0, 10, 20, 30, 40, 50 vol. %
CH_4 fraction in fuel	x_{CH_4}	$1 - x_{\text{H}_2}$
Oxidiser	—	Dry air (21 % O_2 , 79 % N_2)

3.3 Mixture Composition

For each hydrogen fraction x_{H_2} (mole fraction of H_2 in the fuel blend), the stoichiometric oxygen requirement per mole of fuel is:

$$n_{\text{O}_2, \text{stoich}} = 2 x_{\text{CH}_4} + 0.5 x_{\text{H}_2} \quad (1)$$

where $x_{\text{CH}_4} = 1 - x_{\text{H}_2}$. At equivalence ratio ϕ , the actual oxygen supplied is $n_{\text{O}_2} = n_{\text{O}_2, \text{stoich}}/\phi$, and the nitrogen is set by the air composition: $n_{\text{N}_2} = n_{\text{O}_2} \times (79/21)$.

3.4 Grid and Solver Settings

The initial domain width was set to 3 cm. Automatic grid refinement was enabled with the following criteria: ratio 3, slope 0.06, curve 0.12, and a prune threshold of 0.01. These settings ensure that steep gradients in temperature and species profiles are adequately resolved while avoiding unnecessary grid points in smooth regions. The solver was allowed to iterate through three stages of increasing refinement until all criteria were satisfied (auto=True).

4 Results and Discussion

4.1 Laminar Flame Speed

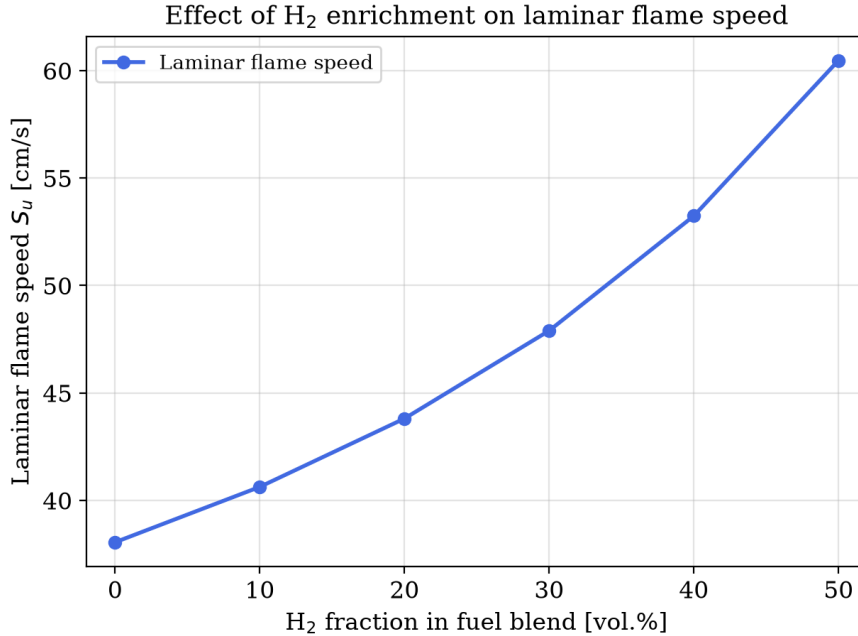


Figure 1: Laminar flame speed S_u as a function of hydrogen fraction in the CH_4/H_2 fuel blend at $\phi = 1.0$, $T_u = 300$ K, $P = 1$ atm.

Figure 1 shows a clear, super-linear increase in the laminar flame speed with hydrogen enrichment. The pure methane flame ($x_{\text{H}_2} = 0$) attains a speed of approximately 37–38 cm/s, consistent with the well-established experimental reference value for CH_4/air at

stoichiometry. As H_2 content rises to 50 vol.%, the flame speed increases by more than 50 %, reaching values above 55 cm/s.

Three physical mechanisms contribute to this acceleration. First, H_2 itself has an intrinsically high burning velocity (≈ 200 cm/s), and even small additions shift the mixture’s effective reactivity. Second, hydrogen’s high thermal diffusivity ($\alpha_{\text{H}_2} \approx 5 \times \alpha_{\text{CH}_4}$) enhances preheat-zone conduction, allowing the unburned gas to reach ignition temperature more rapidly. Third, and most importantly at moderate blending ratios, H_2 dramatically amplifies the radical pool (discussed in Section 4.4), accelerating the rate-limiting chain-branching steps of hydrocarbon oxidation.

The super-linear character of the increase—visible in Fig. 1 as an upward curvature—is a well-known feature of HENG flames and is explained by the synergistic interaction between H_2 kinetics and the methane oxidation network rather than a simple mixing effect.

4.2 Maximum Flame Temperature

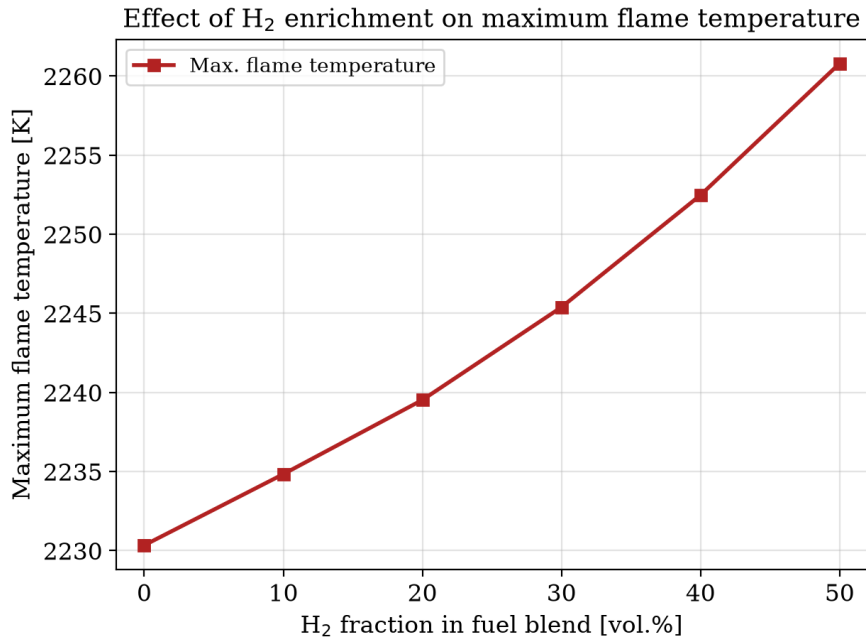


Figure 2: Maximum adiabatic flame temperature as a function of hydrogen fraction.

The maximum flame temperature shown in Figure 2 increases approximately linearly with hydrogen content, rising by roughly 80 K over the studied range. This increase arises from two thermodynamic effects. First, replacing part of the CH_4 (which produces CO_2 and H_2O upon combustion) with H_2 (which produces only H_2O) reduces the specific heat of the product mixture, since CO_2 has a higher molar heat capacity than H_2O . Second, the higher hydrogen-to-carbon ratio means proportionally more water vapour in the products, and the combustion enthalpy released per unit mole of oxygen consumed is slightly higher for H_2 than for CH_4 . Both effects result in a higher adiabatic flame temperature, which has direct consequences for thermal NO formation (see Section 4.3).

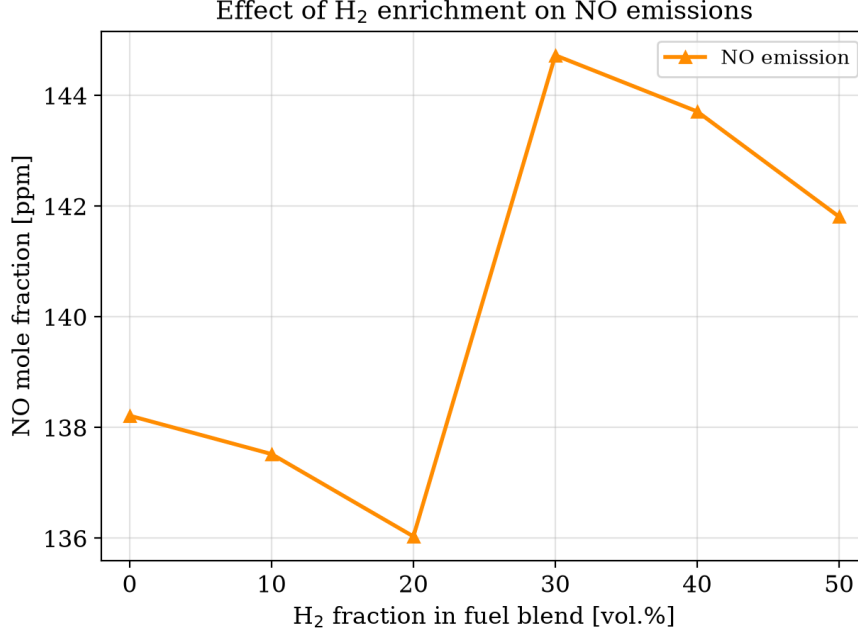


Figure 3: NO mole fraction (as ppm) in the burned gas as a function of hydrogen fraction.

4.3 NO Emissions

Figure 3 demonstrates a significant increase in NO emissions with hydrogen enrichment. The dominant formation pathway in premixed methane/air flames near stoichiometry is the thermal (Zeldovich) mechanism:



The rate of Reaction (2) has an extremely strong temperature dependence (activation energy $\approx 314 \text{ kJ/mol}$), expressed by:

$$k_1 = A T^n \exp\left(-\frac{E_a}{RT}\right) \quad (5)$$

The 80 K increase in T_{max} shown in Fig. 2 therefore translates into a disproportionately large increase in the Zeldovich NO formation rate, explaining the steep rise in NO concentration seen in Fig. 3. The elevated OH radical concentration (Fig. 4) further promotes Reaction (4), creating an additional feedback loop between radical enrichment and NOx formation.

This result underscores a fundamental trade-off in HENG combustion: hydrogen addition improves burning velocity and combustion stability, but simultaneously increases the thermal NOx penalty. Mitigation strategies—such as lean premixing, exhaust gas recirculation, or selective catalytic reduction (SCR)—are therefore critical for practical HENG burner applications.

4.4 Radical Pool Enhancement

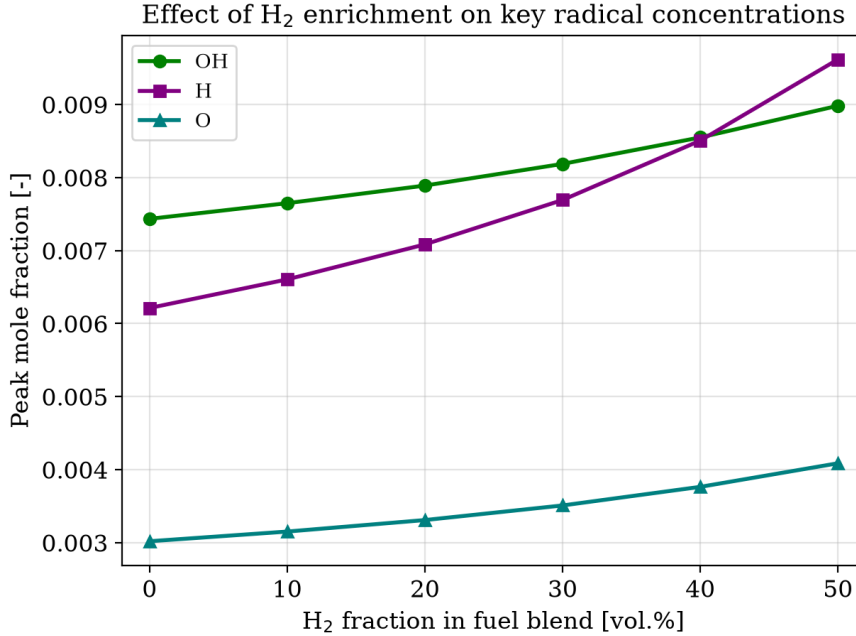


Figure 4: Peak mole fractions of OH, H, and O radicals as a function of hydrogen fraction.

Figure 4 reveals that the peak concentrations of all three key flame radicals—OH, H, and O—increase monotonically with hydrogen enrichment. The H radical, in particular, shows a pronounced increase, consistent with H₂ decomposition acting as an additional source of H atoms in the preheat zone via:



and via abstraction by O:



The resulting surplus of H atoms accelerates the critical chain-branching step:



which simultaneously increases OH and O concentrations. The expanded radical pool accelerates the overall oxidation rate of both CH₄ and CO, leading to the higher laminar flame speeds observed in Fig. 1. This mechanism also explains the super-linear dependence of S_u on H₂ content: the radical chain-branching creates an autocatalytic amplification effect that grows faster than the linear increase in H₂ concentration alone would suggest.

4.5 Spatial Flame Structure

Figures 5 and 6 compare the spatial flame structure for the two extreme cases: pure methane and 50% hydrogen enrichment. Both grids have been shifted so that the point of maximum temperature gradient coincides with $x = 0$, enabling a direct comparison of flame structure rather than absolute flame position.

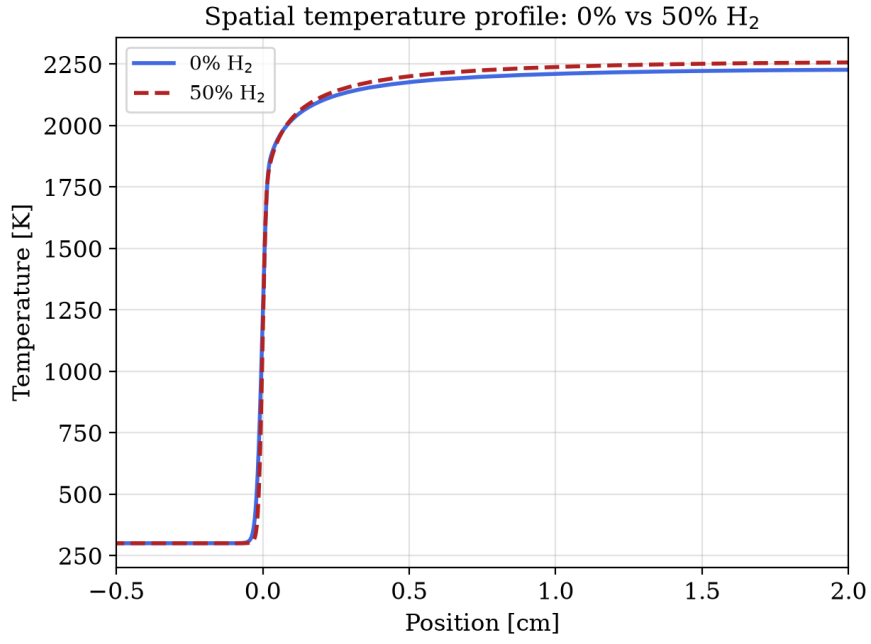


Figure 5: Spatial temperature profiles for the 0 % and 50 % H₂ cases (grids shifted so the point of maximum temperature gradient is at $x = 0$).

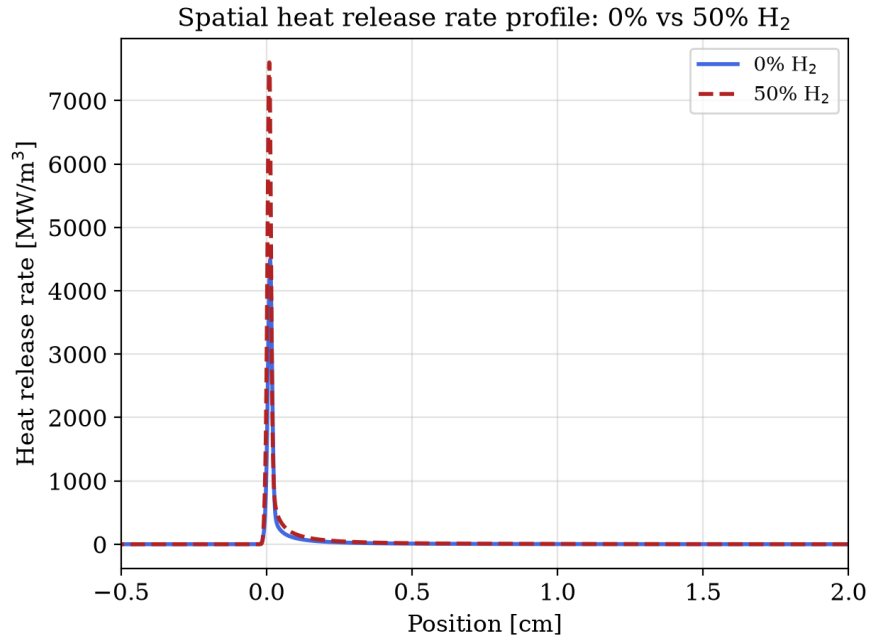


Figure 6: Spatial heat release rate (HRR) profiles for the 0 % and 50 % H₂ cases.

Temperature profile (Fig. 5). The 50 % H_2 flame exhibits a steeper temperature rise over a shorter spatial distance. This narrowing of the preheat zone is a direct consequence of the higher thermal diffusivity of H_2 relative to CH_4 : a larger fraction of the heat released in the reaction zone is conducted upstream into the unburned mixture, raising the local temperature gradient and, by Le Chatelier’s principle applied to the energy equation, compressing the preheat zone. The final burned-gas temperature is also higher, as expected from Fig. 2.

Heat release rate profile (Fig. 6). The peak HRR is significantly higher for the 50 % H_2 case, and the reaction zone (defined by the region where $\text{HRR} > 10\%$ of peak) is markedly narrower. This behaviour is consistent with the higher laminar flame speed: since the flame consumes the reactant mixture faster per unit area, the volumetric heat release rate must increase to supply the required power output. The narrowing of the HRR profile has important implications for turbulent combustion: a thinner reaction zone is more susceptible to turbulent stretching and local quenching, potentially affecting the lean blowout limit at elevated hydrogen fractions.

5 Conclusions

The following key engineering conclusions are drawn from this computational study of hydrogen-enriched methane combustion:

- **Laminar flame speed increases substantially with H_2 addition.** From ≈ 37 cm/s for pure CH_4 , the speed rises to > 55 cm/s at 50 % H_2 , a gain of over 50 % driven by radical-pool kinetics and enhanced thermal diffusivity.
- **Maximum flame temperature rises by ≈ 80 K.** The thermodynamic shift caused by replacing CO_2 -producing methane combustion with H_2O -producing hydrogen combustion increases the adiabatic flame temperature monotonically.
- **NO emissions increase significantly.** The Zeldovich thermal mechanism, highly sensitive to temperature, produces substantially more NO as the flame temperature rises, presenting a critical challenge for emission compliance.
- **Radical pool (OH, H, O) is enriched by H_2 .** The chain-branching cycle $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ is strongly amplified by the additional H atoms provided by H_2 decomposition, explaining the super-linear increase in flame speed.
- **The reaction zone narrows and intensifies with H_2 addition.** Spatial profiles show a thinner preheat zone and a higher, narrower HRR peak, implying increased susceptibility to turbulent disruption and flashback at high H_2 fractions.
- **Hydrogen blending is a viable but constrained decarbonisation strategy.** While HENG reduces CO_2 emissions proportionally to the hydrogen fraction, the NO_x and flashback penalties require active mitigation through burner redesign, lean premixing, or post-combustion treatment.

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